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Environment-framed networks: seasonal reconfiguration of service accessibility in Arctic transport systems

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Abstract

While transport accessibility is an integral part of quality of life, its value does not remain constant over time. In practice, access to services can be crucial, especially in the case of medical services and transport hubs. This is particularly evident in remote regions such as the Arctic, where settlements are highly interdependent. In harsh climatic conditions, transport links become vulnerable to disruption, and strong seasonal fluctuations significantly alter the structure of the network. On the other hand, in the Arctic, unlike in more temperate climates, the changing seasons also bring new opportunities, such as ice roads. Traditional frameworks of service accessibility assessment, however, rarely take into account such climate-induced network reconfiguration. To study this phenomenon in detail, we propose an Environment-Framed Network (EFN) model for analyzing seasonal resilience of service accessibility in Arctic settlement systems under climate-constrained transport connectivity. EFN represents transport as a temperature-dependent backbone and models network layers as service-specific population flows, enabling explicit treatment of environment-driven link existence. Simulations across four Arctic regions reveal threshold-driven seasonal reconfigurations: up to 40% of settlements switch service providers, and 15–20% become temporarily isolated for up to 2–4 months during seasonal transitions. Despite strong regional heterogeneity, these dynamics are consistent and governed by the instability of seasonal transport modes. The results show that resilience in climate-dependent regions is determined primarily by the stability of service catchments, rather than by static network connectivity.

Keywords Complex networks, Urban agglomeration, Urban resilience, Arctic

Introduction

As one of the most sensitive regions to climate change, the Arctic is undergoing rapid transformations that are having a profound impact on transportation systems. Rising air temperatures, thawing permafrost, and shrinking ice cover create challenging conditions for the operation of both land and sea routes Dong et al. (2025); Stephenson et al. (2011). In recent years, researchers have noted a paradoxical dynamic: while shipping in the Arctic is becoming more accessible due to the decreasing extent of sea ice, land

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transportation infrastructure, including winter roads and airports, faces increasing risks due to degradation of the permafrost and decreasing periods of sustained low temperatures (Stephenson et al. 2011; Waite et al. 2023).

Settlement agglomerations/systems management challenges

The accessibility of transportation in the Arctic is affected by a broad set of climatic parameters, among which the most significant are surface air temperature, permafrost condition, and ice cover dynamics (Dong et al. 2025; Waite et al. 2023; Wang et al. 2020). The increase in average annual temperatures leads to a reduction in the period during which the construction and operation of winter roads is possible. Studies show that this process is faster in North America than in Eurasia, where colder winters still allow ice crossings to remain longer (Dong et al. 2025; Povoroznyuk et al. 2022). However, even in the Eurasian part of the Arctic, there is a steady trend towards a decrease in the number of days suitable for the use of such roads (Dong et al. 2025; Waite et al. 2023; Povoroznyuk et al. 2022). The transportation network plays a key role in ensuring that people have access to essential services. However, the road network often provides uneven levels of accessibility for different communities. In addition, the accessibility of critical facilities is constantly interrupted by emergencies (Gangwal et al. 2023).

A particular threat is the degradation of permafrost, which serves as a natural base for many infrastructure facilities. Warming leads to an increase in the depth of seasonal thawing, which involves ground subsidence, road bed failure, and runway deformation (Waite et al. 2023; Wang et al. 2020; Mahmoud et al. 2024). These processes require significant repair and modernization costs, which is especially problematic in remote and sparsely populated regions with limited financial resources (Stephenson et al. 2011; Waite et al. 2023; Coates and Broderstad 2019).

One of the key conclusions of the reviewed studies is the dual impact of climate change on the Arctic transportation system. On the one hand, the reduction of sea ice opens up new opportunities for shipping (Stephenson et al. 2011; Mahmoud et al. 2024). On the other hand, land-based transportation infrastructure is facing increasing challenges. Winter roads, which traditionally served as the main means of communication between remote settlements during the cold season, are becoming less reliable due to the shorter period of sustained frost (Dong et al. 2025; Stephenson et al. 2011; Coates and Broderstad 2019). As a result, many regions may face increased transport isolation, especially in the spring and fall periods, when the use of both conventional and winter roads becomes impossible (Dong et al. 2025; Waite et al. 2023).

Settlement systems management technologies

In general, modern research in the field of urban sustainability shows significant progress in the development of methodological approaches and indicator systems to assess the sustainability of the transportation system. The reviewed studies have evolved from basic statistical methods to advanced hybrid algorithms. In addition to considering the effect of temperature, some studies consider temporal changes. The study (Huang et al. 2022) examines spatial and temporal changes in the sustainability of ecological networks (EN), focusing on the landscape and the dynamics of the ecosystem in the evaluations. It uses chain theory to map the migration paths of organisms and identify barriers to the corridor, as in the study (Wang et al. 2023). In (Lu et al. 2022), genetic algorithms are

combined with neural networks, enabling optimal solutions under high uncertainty typical of urban systems. Paper (Sun et al. 2024) introduces the CRITIC entropy method, which balances indicator significance by contrast and information richness, reducing subjective bias in weight determination. The network method is one of the most appropriate and frequently used approaches for systematically assessing the availability of critical facilities (Gangwal et al. 2023). By looking at a set of social indicators, such as membership of different communities or household composition, the authors examine the effects of flooding. They used a 100-year flood map to determine which roads would be flooded. This approach is also relevant to the Arctic, where predicting long-term changes is critical to the operation and planning of transportation infrastructure. However, digitalization and environmental analytics remain missing (Huang et al. 2022; Wang et al. 2023; Lu et al. 2022; Ji et al. 2024).

Problem statement

The most significant gap in the analyzed works is the insufficient consideration of climatic features and their impact on urban systems in the long term. For example, the indicators of transport accessibility, discussed in detail in (Yin et al. 2023), do not take into account that in winter many roads in northern regions become difficult to pass or are closed altogether. Similarly, the environmental indicators from (Wang et al. 2023) are not adapted to assess the impacts of extreme weather events, which are becoming more frequent due to climate change. In addition, because of significant climate change, the Arctic requires consideration of impacts in a few decades, as design solutions for northern roads should be based on projections of changes in permafrost conditions for at least 50 years after construction (Povoroznyuk et al. 2022).

Current urban resilience research provides powerful analytical tools but needs significant refinement to account for temporal and climatic factors. Special attention should be paid to Arctic territories, where traditional resilience assessment methods are often inadequate due to extreme climatic conditions (Coates and Broderstad 2019). Specialized approaches are required here, taking into account such factors as permafrost, polar nights and extremely low temperatures (Lowe and Sharp 2021).

Hypotheses

This study develops an Environment-Framed Network (EFN) model to test the following hypotheses:

- H1: Remote urban agglomerations exposed to climatic forcing exhibit similar changes in seasonal accessibility patterns.
- H2: Seasonal climatic transitions induce systematic service-provider switching and temporary service unavailability.

Unlike classical multilayer transport networks, EFN treats transport modes as a shared, environment-constrained backbone and defines layers as service-specific flows over that backbone.

Related works

In this study, the terms urban agglomerations, settlement systems, and networks of settlements are used interchangeably to denote a functionally connected set of cities and rural settlements linked by shared transport infrastructure and service accessibility relations.

Network resilience

Overall, network resilience theory provides a unified perspective for complex systems in biology, infrastructure, and socioecology. Its core ideas—phase transitions in interconnected systems, recovery based on centrality, and multilevel vulnerabilities—offer practical principles for improving resilience.

Network resilience has been studied in various fields and research areas, from climate to transportation. Examples of some of these studies include climate networks (e.g., moisture recycling in the Amazon), which demonstrate feedforward loops that accelerate cascading transitions. Several models have been proposed for analyzing network resilience. Regional resilience networks, such as China's economic zones, use modified gravity models to quantify resource flows. Cities with high interconnectedness recover faster but face higher risks of cascading effects during financial crises (Wang et al. 2023).

Urban and agglomeration structural stability

In recent years, the resilience of urban and agglomeration (or regional) networks has been widely studied (Jie et al. 2024). Research shows that both individual settlements and the entire network of cities suffer simultaneously, experiencing cascading failures as a result of disruptions. The losses incurred can be viewed from an economic, functional, or topological perspective (Yin et al. 2023). The phenomenon of urban resilience is examined through specific areas, such as economic or infrastructural urban resilience (which, in addition to other parameters such as road density, includes key urban services such as hospitals) (Guo et al. 2024); and also link it to areas of organizational sustainability (market, society) and urban network sustainability (population network, innovation network, ecological network) (Mu et al. 2022; Huang et al. 2022), which can be analyzed separately or averaged as a combined measure of urban sustainability (Ji et al. 2024). They also highlight the limitations of agglomeration-level research in terms of urban network sustainability, which is important given the interdependence of nodes in the city network.

Urban resilience measures

(Jie et al. 2024) lists a set of parameters used in research, mainly to assess the resilience and reliability of the urban environment. Other works also include indicators of vulnerability and anti-fragility (Kiparisov and Lagutov 2024). To assess the resilience of a system, the authors mention the level of hierarchy and assortativity coefficient, degree distribution, and degree correlation. Reliability is explained by the level of structural redundancy (density, average number of independent paths, network structure entropy), modularity, hierarchy, and transmission indicators. In another line of research, the resilience of cities (in the context of disruptive events such as natural disasters) is viewed as the resilience of communities to the nonlinearity of complex urban dynamics, with resourcefulness added as an element of resilience. The study of urban subsystems adds a

quantitative assessment of resourcefulness to weigh the aforementioned characteristics against other indicators (e.g., considering not only the number of base stations but also internet speed) (Yin and Mostafavi 2023).

Temporal connectivity resilience of network of cities

Cities are connected by transport networks not only by roads, but also by air and water transport. As shown earlier, connectivity, as reflected by various network attributes, is crucial for the stability of network functioning. Although changes in urban networks are often viewed as rapid changes in their properties as a result of specific events, such as natural disasters, traffic accidents, etc. (Yin et al. 2023). Another example of long-term changes is the spatiotemporal evolution of the network and its resilience properties, as assessed for an ecological network (EN) in (Huang et al. 2022). Changes in EN parameters, including source and corridor (edge) quality, were studied at 8-10 year intervals from 2000 to 2018, revealing significant changes in the variables under study.

Distinction of the proposed model (EFN)

The layers are service flows rather than transport subnetworks. Unlike most works on multilayer transport networks, where the layers are different transport subsystems (aviation, rail, road, water transport, and their connectivity and sustainability; see, e.g. (Li et al. 2025; Mei et al. 2025; Ye et al. 2025)), in EFN, layers are defined by flows of movement between cities in terms of a specific service. Settlements have different sets of services and attract populations from settlements that do not have these services in different ways. Thus, the set of network layers is formed precisely as a set of flows to different types of services, rather than as a set of infrastructure subsystems. This brings EFN closer to the literature on trip distribution and the assessment of social facility provision (Mishina et al. 2024), but shifts the focus to a seasonally changing, climate-constrained network of connections.

In EFN, transport links are not layers of the model; they are represented as edges of a single backbone network that can belong to different modalities (land, water, air) and be subject to different constraints. Traditional network resilience studies often analyze the importance of nodes (hubs) and the multi-layer effects of their damage (Mei et al. 2025; Ye et al. 2025), but in EFN, the nodes are real settlements, for which short-term “destruction” as a reaction to external disturbances is uncharacteristic and can only be considered in the long term (population outflow, closure of production facilities).

At the same time, flows pass along edges — transport links of different modalities, which are affected by weather and climate factors. Changes in temperature and weather conditions can simultaneously affect several modes of transport, as discussed in detail in systematic reviews on weather disruptions in transport and logistics (Touloumidis et al. 2025), on the resilience of bus networks in cold regions (Gu et al. 2025), the adaptation of the aviation sector to climate change (Burbidge et al. 2024), and summaries of climate risks and their consideration in infrastructure planning (Rahman et al. 2025; Food 2010). In EFN, it is precisely these climate-dependent changes in the state of the edges that lead to a redistribution of flows and, ultimately, to changes in the level of accessibility of social infrastructure in large cities in the agglomeration.

EFN dynamics manifest themselves in seasonal changes in transport connections depending on limiting factors (temperature, weather conditions). The evolution of the

network occurs due to (i) changes in the set of edges (their existence/non-existence) and (ii) the redistribution of demand between nodes. Layers, interpreted as population flows to specific services, are independent in the current setting and do not generate cross-level cascading effects. This distinguishes EFN from works that explicitly model cascading failures in multilayer transport networks (e.g., in metro-bus dual networks under the influence of temporarily changing passenger flows (Guo et al. 2025; Rouhana and Jawad 2025), or in inter-level road networks with cross-layer route selection (Ye et al. 2025).

The authors agree that differentiating the consequences of failure of nodes with different structural significance (for example, a regional center versus a peripheral village) is an extremely important aspect of network stability analysis. In this paper, we focus on another systemic effect — the dynamics of the redistribution of availability flows between multiple nodes in conditions of seasonal disturbances, where the “disappearance” of any settlement simulates the loss of a local service delivery center. The collapse of a large transport and logistics hub would have catastrophic consequences that go beyond the seasonal dynamics under consideration and lead, among other things, to long-term migrations and the desolation of dependent territories. Such node-based disruptions are not the focus of this work.

The study of the resilience of precisely such supporting, highly centralized nodes to extreme climatic or infrastructural disruptions requires separate, more detailed modeling with different initial assumptions and is the subject of our further research program.

However, as the results show, when considering the dynamics of flow distribution in EFN in a comprehensive manner, the areas of interaction between cities change: areas of attraction to different centers are reformatted, which allows analyzing the seasonal evolution of “service areas” depending on the state of the environment and the network.

EFN emphasizes the importance of meeting demand for services, as access to them directly affects quality of life, especially in remote settlements where the consequences of insufficient accessibility are most critical. Here, EFN develops a line of work on assessing the provision and accessibility of social infrastructure (Mishina et al. 2024), but considers it in the context of explicitly modeled external constraints (climate, seasonality), embedding it in the broader context of climate-oriented solutions for infrastructure and supply chains (Rahman et al. 2025).

Transport modes and environmental constraints in the Arctic

The transport infrastructure in Arctic regions around the world is characterized by a linear nodal structure with a low network density (Dankin et al. 2022). This is due to extreme climatic conditions, low population density and the complexity of construction in permafrost conditions. Most territories do not have year-round transport links, and the available modes of transport (mainly aviation and water) are characterized by high operating costs. At the same time, there are significant differences in transport accessibility between more developed coastal areas and remote inland territories.

Table 1 summarizes the main modes of Arctic transport, their operating temperature ranges, and environmental thresholds. In this paper we mainly focus on operating temperature range as a key constraint.

For agglomerations, as for cities, time-to-service duration constraints were also applied, which differs between regions. Thus, even if a route exists, it should pass accessibility constraints to be further considered as an active edge in the network.

Table 1 Operational constraints for Arctic transport modes

Mode	Operating temperature (°C)	Constraints
Automobile	[−60; +40]	Snow drifts >50 cm, visibility <50 m (blizzard) (Ministry 2020; Nordic 2020; Finnish 2019)
Small aircraft (AN-2, MI-8)	[−50; +45]	Wind >15 m/s, runway icing, visibility <1000 m (Various 2009; Ministry 2009; Kong and Doré 2024; Transport 2024)
Water transport (boats, ferries)	[−2; +45]	Waves >4 pts, ice thickness >15 cm, storms (Government 2020; International 2014; International 2017)
Snow vehicles (e.g., snowmobiles)	[−50; +10]	Loose snow >1 m (Federal 2017; Povoroznyuk et al. 2023; Barrette et al. 2022)

Arctic unique challenges

The Arctic regions are a unique and extremely complex object for network analysis due to the combination of extreme natural, infrastructural, and socio-economic factors that form a fundamentally different logic of transport links compared to temperate latitudes.

Firstly, classical static networks (highways, railways) are fragmented or absent here, giving way to dynamic, seasonal and informal communication routes: winter roads that exist only a few months a year; temporary ice crossings; snowmobile trails, the routes of which vary depending on weather conditions. Secondly, multimodality acquires not an additional, but a mandatory and life-supporting character, where the same route between two points can sequentially include moving by car through a winter road, reloading onto a boat to overcome a water barrier and ending with a helicopter section. Thirdly, data on such links are often unsystematic, local in nature, or exist only in the form of oral practice, which makes traditional sources (such as OpenStreetMap (Haklay and Weber 2008)) incomplete and requiring complex verification.

Finally, the spatial organization of Arctic communities often follows the logic of access to resources (deer pastures, fishing grounds, oil and gas fields) rather than the principles of economic optimization, creating network structures poorly described by models developed for urbanized territories. Thus, the network model for the Arctic is not a fixed infrastructure, but a dynamic, multimodal and contextually dependent system, the analysis of which requires special methodological adaptations to account for its fundamental instability and weak formalization.

Modes and seasonal feasibility

- **Road transport** is key in winter when rivers freeze into seasonal ice roads. However, it fails in deep frost (below −50°C) and spring thaw.
- **Small aircraft** ensure year-round access in isolated areas but are vulnerable to icing, storms, and low visibility.
- **Water transport** dominates in summer, but is limited by the short navigation season (2–4 months), ice coverage, and storm threats.
- **Specialized vehicles** (snowmobiles, all-terrain) are widely used by indigenous groups and industry. However, permafrost melting affects terrain passability.
- **Railways** are almost absent due to extreme construction and maintenance costs.

Common Arctic transport challenges

The transport systems of all Arctic regions of the world face a common set of challenges caused by extreme natural conditions (Dankin et al. 2022; Serova and Serova 2019). All the northern territories are characterized by:

- **Seasonality** — many links are only viable for part of the year (e.g., winter roads, water routes).
- **High weather dependence** — strongly affecting reliability and planning.
- **Linear, disconnected networks** — settlements link to regional hubs, but not to each other.
- **Multimodality** — transport switching between seasons is mandatory.
- **Extremely high costs** — infrastructure and operation often exceed 5–10 times temperate costs.

These conditions are found across the Arctic. For example:

- **Scandinavia** has the most robust road network, enhanced by ferry links.
- **Canada, Russia, USA** depend on aircraft and seasonal roads.
- **Greenland** lacks road network, fully dependent on sea and air.

Data

Data sources

For this study, we mainly used OpenStreetMap (Haklay and Weber 2008) as a data source for road and water infrastructure, cities and regions borders. To obtain existing transport connections among remote settlements we searched local websites, social media groups and interviewed local residents where it was possible. Transport times were collected from local transit websites and regional transport agencies. The actual list of data sources is provided in A. The temperature is obtained using NASA POWER API (Nasa Power Project 2024) with a spatial resolution of temperature at 2 m of 0.5×0.625 Degrees, providing temperature data since 1982.

Data limitations

The key data limitation of this study, which also touches the topic of reuse of the proposed method is a scarcity of input data. Being well established in large cities, open transport data is very limited in remote regions, especially for small remote settlements. Another important topic which needs to be covered is a network parameters. Usually in network science the network disruption is considered as a situation when the network disassembles to the extent then one or more nodes stop being connected with the rest of the network. However, based on conducted experiments, for the network of Arctic cities in some areas this is a common situation. Thus in such cases we consider the change in number of months while certain city dose not have an access (in a specified time constraint) to the desired service.

Data schema: graph

Network

Networks themselves are represented as groups of connected cities in an Arctic region, connected by different kinds of transport modes. Initial state of the network is a cold period (which is most probable period for the Arctic region), dated back to the early 1980-s, the beginning of measured temperature time period. The states of the network evolves monthly until 2025th. Thus, each year the Network is represented by 4 seasons of different durations. However the Network states are considered not by seasons but through the limit temperatures, which exceed the threshold of the existence of certain transport mode. Initially every network doesn't have islands, has a few clear hubs and in most cases is represented as a MultiGraph data class.

Edges

In this model links are simple connections among cities, representing time (or distance) among two geographical points. Graph of urban agglomeration might have several edges among the same nodes since there are more than one transport option to reach one city from another (such as private car, boat, plane). Each link has transport mode and time as a weight of the edge. However it is important to note that in such remote area as an Arctic region, there is a seasonal specific regarding routes for warm, cold and middle seasons. Thus, there are a few additional types of transport modes, representing the existence of the link. Basically, the private car mode splits into two specific periods (warm and cold), apart from the ones existing through all the seasons. In the midseason it is considered that none of the latter exist. Warm period roads represent less popular routes that exist only in good environmental condition due to the difficulty maintaining them through the year. The roads of the cold period exist in specific cold temperatures. When there is enough snow, it is compressed via specific machines to create road-like surface out of the snow. Such routes are unique and might not repeat the existing warm period routes, but create new edges, connecting cities only in this time of the year.

Nodes

In this work, nodes represent towns and villages. Each settlement has a set of following parameters: population, number of services of selected types, list of transport facilities such as marinas or airports.

Study area

Due to data availability in this research, we consider urban agglomerations in Eurasian Arctic. Some of the specific routes were obtained from local websites and other very area-specific web resources, local communities, and news feeds. Although the method itself remains to be scalable due to general requirements for the input data (Fig. 1).

It is clear that the spatial distribution of nodes of the same color is wide. Such remote nodes, such those in the orange group, represent large cities, which play an important role for the main group of nodes, being important logistics hubs due to strong aviation connectivity.

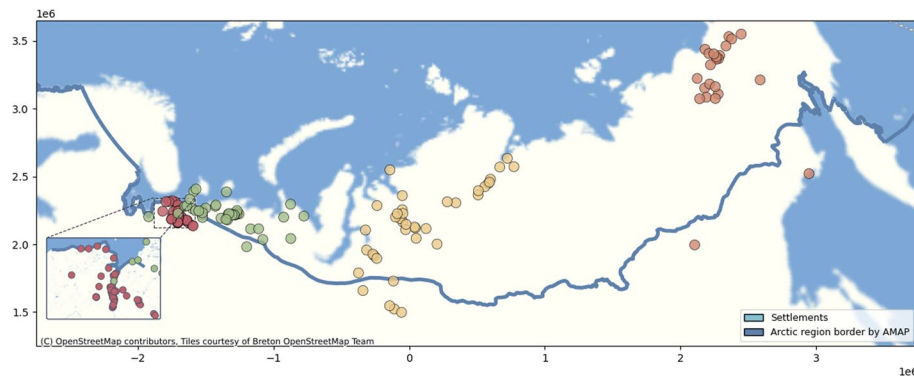


Fig. 1 Study area map. Settlements in agglomerations are differentiated by colors: red – Mezen, green – Northern Administrative District, yellow – Yamalo-Nenets Autonomous District, orange – Yakutia/Chukotka

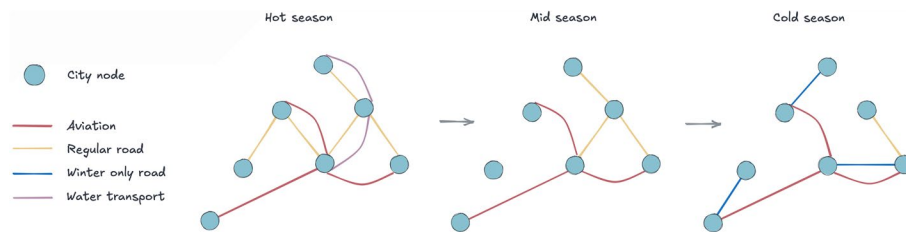


Fig. 2 Network scheme

Modeling

Model proposed in the article

In this article we study the network of cities in areas, where the seasonal influence of climatic changes is strong and affects the connectivity between cities. To model and predict evolving seasonal patterns, we present the environment-framed network model (EFN). EFN model has two parameters: the global external parameter of the environment temperature T , the internal set of parameters of the model such as the graph connectivity parameter for a graph $G(t)$, which spans the nodes (cities). The model allows to describe how the connectivity of $G(t)$ depends on the environmental parameters, such as temperature.

We test the EFN model on the resilience properties looking at: deletion of OD pairs based on the connection probability threshold (Berthelot et al. 2020); stability of the network based on the fulfilled demand for selected services (e.g. clinic in desired time accessibility).

Environment-framed (temporal) network model

In the EFN framework, the transport network topology evolves through temperature-dependent edge feasibility, while service accessibility evolves through reassignment of OD flows over the feasible network. As a result, temporal changes are analyzed at two coupled levels: (i) network connectivity and (ii) provider-specific service catchments derived from flow assignments (Fig. 2).

We consider a dynamic graph $G(t) = (V, E(t))$, where edges exist probabilistically depending on time and temperature.

Edge existence modelling.

Let k be the type of connection:

- k_1 : Regular road
- k_2 : Aviation
- k_3 : Winter only road
- k_4 : Water transport

The feasibility of a transport connection of mode k between settlements i and j at time t is determined exclusively by temperature and is quantified by a temperature-dependent feasibility function:

$$P_{ij}^k(t) = \mathbb{P}(E_{ij}^k(t) = 1 \mid \tilde{T}_{ij}(t)) = \frac{1}{1 + \exp\left(-\frac{\tilde{T}_{ij}(t) - \theta_k}{\beta_k}\right)}. \quad (1)$$

Here, $\tilde{T}_{ij}(t)$ is the effective temperature proxy for the origin–destination pair (i, j) , θ_k denotes the critical temperature threshold associated with transport mode k , and β_k controls the steepness of the transition between infeasible and feasible regimes (see D for detailed explanation).

$P_{ij}^k(t)$ is interpreted as a *feasibility weight* that determines whether a transport edge of mode k is available at time t . It does not represent a service accessibility condition.

Service accessibility

Service accessibility is evaluated *after* transport feasibility has been determined. For each region r , we define a maximum admissible travel time D_r reflecting regional planning norms.

A settlement i is considered connected to a service provider j at time t if there exists at least one feasible path composed of available transport edges such that the corresponding travel time $\tau_{ij}(t)$ satisfies

$$\tau_{ij}(t) \leq D_r.$$

If no provider satisfies this condition, the settlement is considered isolated for that service at time t .

Region-specific values of D_r are taken from official planning regulations (Table 2).

For each service m (e.g., port m_1 , hospital m_2), we define

$$\delta_i^m = \begin{cases} 1, & \text{if service } m \text{ is present at node } i, \\ 0, & \text{otherwise.} \end{cases}$$

In summary, temperature controls transport feasibility (edge existence), whereas travel-time thresholds control service accessibility over the feasible network.

Table 2 Time accessibility constraints for regions

Region	Time accessibility constraint
Yakutia/Chukotka	3 h (Department 2022; Government of 2011)
Yamalo-Nenets Autonomous District	1.5 h (Government 2018; Government of and Krai 2014)
Mezen	1 h (Government 2016)
Northern Administrative District	3 h (Department 2024)

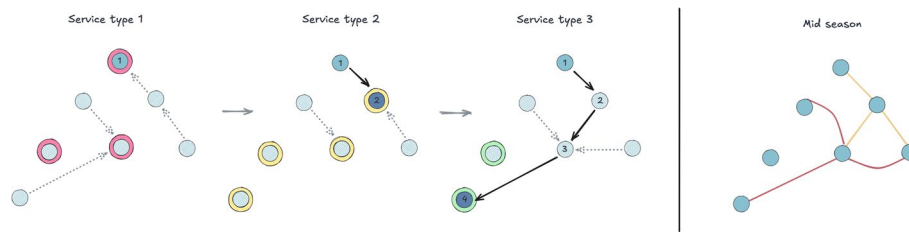


Fig. 3 Flow redistribution scheme for transitional season (autumn / spring)

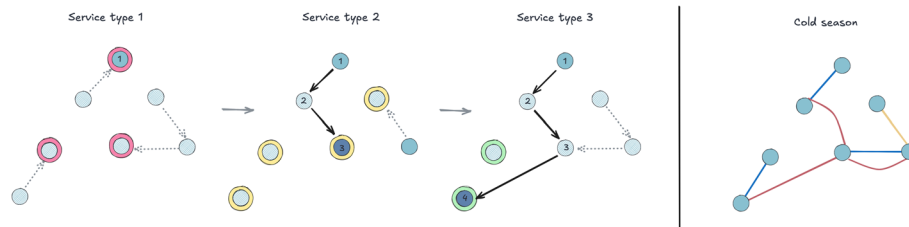


Fig. 4 Flow redistribution scheme for cold season (winter)

Figures 3 and 4 display schematic version of flow redistribution across different levels of network (service layers) in warm and cold periods. Each figure displays nodes (settlements) with services with colored stroke, where color reflects service type. Schematics represent flows by directed edges. Since the examples tailored to showcase seasonal changes for the Node 1, bold edges represent its flow to the destination node. Here, edge existence displays both physical existence and fit to regional accessibility norms.

Results

Results are reported at three complementary scales: network-level connectivity, service-specific flow assignments, and region-level aggregates.

Network-level seasonal dynamics

The simulation results show that urban agglomerations in the considered Arctic region display different fragility patterns. Differences in the distribution of key (serving) cities, the diversity of transport modes, and the high dependency of small towns on seasonally unstable transport modes (such as water transport or winter roads).

This network instability caused by seasonal temperature changes leads to untracked changes in consumer-provider relationships. Some settlements might not be connected to a provider for a period of several months due to missing transport connections, which is somewhat normal for remote towns and villages in such regions.

In addition, the urban agglomeration network, as in most urban networks, is a multilayer network. The nodes (settlements) interact differently on each of many possible levels. In the context of consumer-provider relationships, uneven distribution of services creates complex relationships. This example can be seen in 5.

The existing edges at 5 also differ in the plots for different months. Gray links show whether the link exists according to the temperature and transport type rule, and colored links show demand routes, displaying the potential flow from one settlement to another, according to service demand.

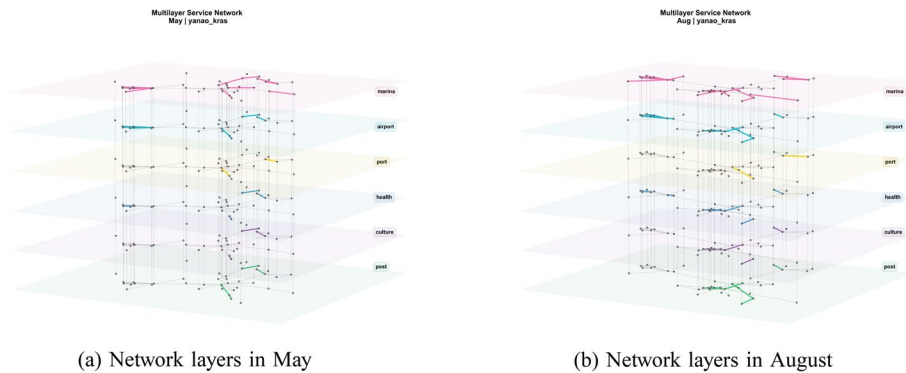


Fig. 5 Seasonal changes in multilayer network interactions between May and August for Yamalo-Nenets Autonomous District

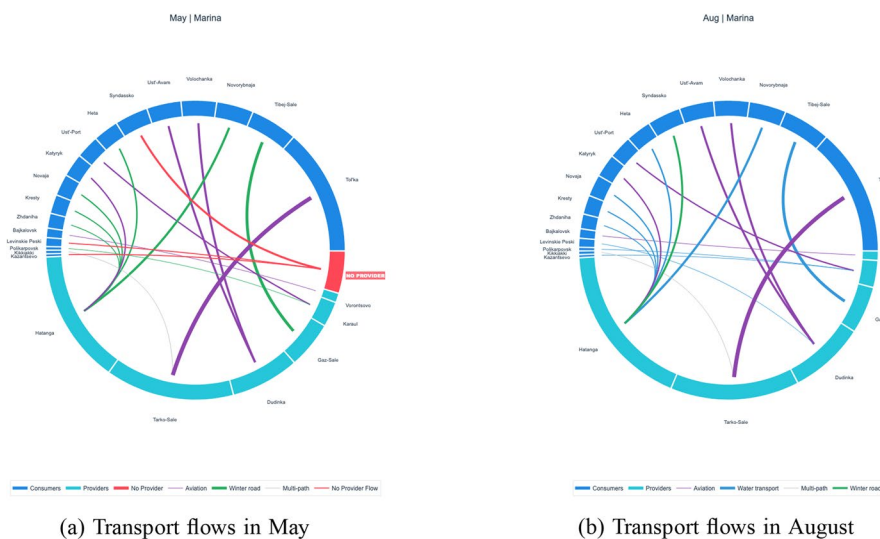


Fig. 6 Seasonal changes in routes by transport type between May and August in Yamalo-Nenets Autonomous District

Flow and provider reassignment

In Fig. 6 it can be clearly seen that certain small settlements (presented on the left side of the circled chart) are not connected to any services in a specified temporal threshold. While on the right chart (presenting network state 3 months after) all the settlements have an access to the desired service (marina in this case). It is also important to highlight that several settlements changed the type of transport connection. This can be studied on the example of Heta village. First, in May it was connected to the marina service via winter road, but in August the Water transport started to play main role connecting Heta to the nearest marina. However, as presented in Fig. 7, besides changing the transport type, the settlement remained to be connected to the same provider.

Catchment (assignment/ isolation) stability, similarity and evolution

For each month, settlements are grouped by their assigned service provider, forming provider-specific service catchments. These catchments represent assignment-based allocation basins rather than topological communities detected from network structure.

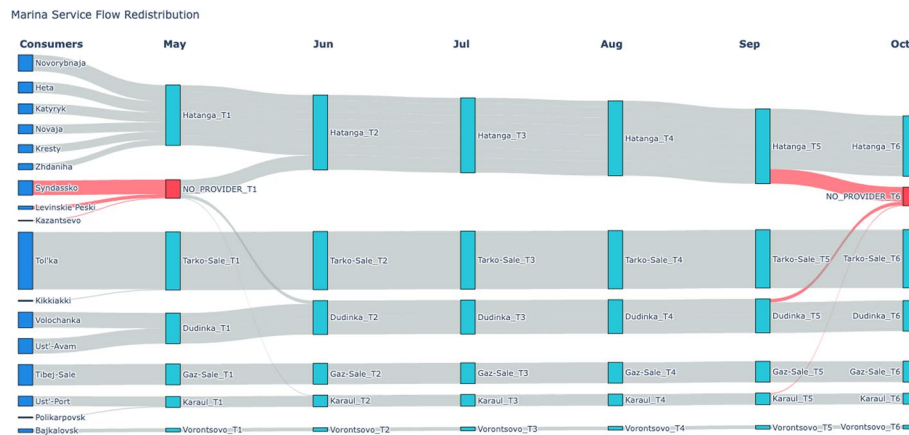


Fig. 7 Flows redistribution in Yamalo-Nenets Autonomous District

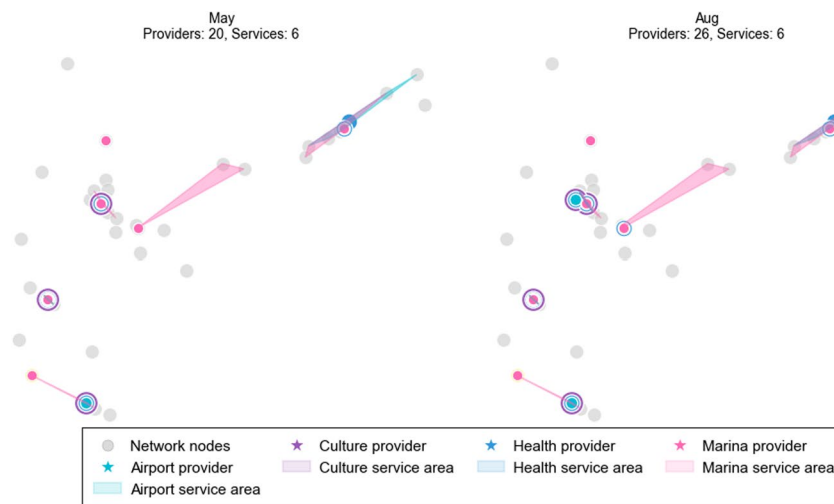


Fig. 8 Service area changes in Yamalo-Nenets Autonomous District

Temporal similarity between consecutive months is quantified using the Jaccard index of settlement sets.

In particular, in Fig. 6 can be seen that the access routes for the marina (second blue layer in Fig. 5) change significantly. This case shows routes instability and fragility of the access from affected settlements. As will be shown later in Fig. 10, being unstable in season transition months (May, September), in between periods, the overall picture seems more or less consistent.

Despite transportation type changes due to seasonality only a few settlements lose access to the selected service. Although not stable in global terms, the system regains its initial state in October, indicating clear state transitions in the mentioned period.

Service catchments structure change over time

Providers change mentioned in Fig. 7, occurring due to seasonal transport changes, shown in Fig. 6 both indicate displacement of service areas. This evolution could be represented as a who-serves-whom matrix, presented as a polygonal service area map in Figs. 8 and 9. It can be seen that there are changes in both service areas and active

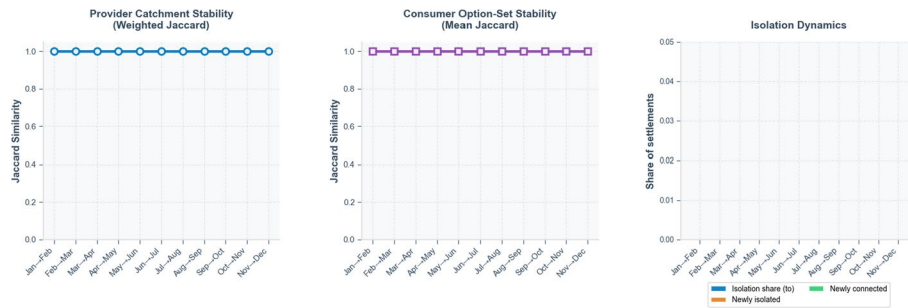


Fig. 9 Service catchments evolution in Mezen for Marina service

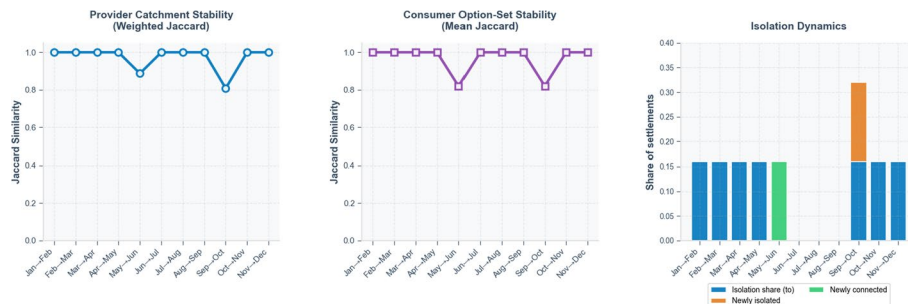


Fig. 10 Service catchments evolution in Yamalo-Nenets Autonomous District for Marina service



Fig. 11 Service catchments evolution in Northern Administrative District for Marina service

providers for different services. Nodes are colored by the type of facility they serve access to.

Seasonal structural changes in the system of Arctic urban agglomeration network take place during certain months, during which the transition occurs due to temperature changes. Month-to-month similarity of service catchments was quantified using the Jaccard index, computed as the ratio of intersection to union of settlement sets, represented on service catchments similarity plot in Fig. 10.

Although the service-provider relationships appear to be more consistent in a warm period, which is indicated by normalized mutual information plot in the Fig. 10. Service catchments evolution events plot in Fig. 10 highlights the different patterns of transitioning from one state to another from May-to-June to September-to-October. Transitioning from cold to warm period requires only one step, while the reverse transition takes two steps (from 8 to 10 service areas and back) (Figs. 9, 10, 11 and 12).



Fig. 12 Service catchments evolution in Yakutia/Chukotka for Marina service

Table 3 Isolation persistence, provider switching, and network structure metrics

Metric	Yakutia/Chukotka	Yamalo-Nenets Autonomous District	Mezen	Northern Administrative District
Number of settlements	23	38	47	44
Mean isolated months	9.095	3.263	9.525	1.773
IQR isolated months	1.0	8.0	1.0	0.0
Peak isolation share	0.762	0.289	0.800	0.182
Peak switch share	0.143	0.079	0.000	0.432
Peak switch months	May–Jun; Sep–Oct	May–Jun; Sep–Oct	Jan–Feb to May–Jun	Jun–Jul; Jul–Aug
Peak isolation months	Jan–Jun; Aug–Oct	Jan–May; Oct–Dec	Jan–Jun; Aug–Oct	May–Jun; Aug–Sep
Density	0.464	0.126	0.255	0.179
Average degree	9.278	4.667	9.942	7.689
Clustering coefficient	0.567	0.459	0.579	0.756
Diameter	3.167	6.667	8.000	4.000
Modularity	0.022	0.413	0.263	0.233
Number of communities	7.33	9.33	4.00	8.17

Isolation metrics are computed from monthly service availability. Network metrics are averaged over monthly graphs. Modularity and number of communities are obtained using greedy modularity optimization

Regional regime comparison

The considered regions differ by location, spatial distribution and density of land, administrative borders, and the amount of big cities with close access 3. Although being in an Arctic region they show different state transition patterns.

Across all four regions (Mezen, NAO, Yamalo–Nenets AD, Yakutia/Chukotka), accessibility dynamics follow a common seasonal regime characterized by: sharp degradation during spring and autumn transition months, partial stabilization during cold and warm plateaus, recovery to structurally similar states after transitions.

Synchronized drops in Jaccard similarity during May–June and September–October consistent timing of peak isolation and provider switching months across regions, repeated emergence–disappearance cycles of service catchments rather than monotonic degradation.

At the same time, regional topology modulates severity, not mechanism. For example, Yakutia/Chukotka exhibits longer isolation persistence due to sparse topology and winter-road dependence, while NAO shows shorter but sharper switching episodes linked to aviation–water substitution. This confirms mechanistic universality with structural heterogeneity, directly supporting the first hypothesis.

The results show that seasonal transitions are not marginal perturbations but system-level reconfiguration events: switching and isolation peaks are temporally aligned with temperature threshold crossings, not random failures. It is demonstrated that provider switching is driven by loss of feasibility of entire transport modes, forcing reassignment of OD flows even when settlements themselves remain intact.

Discussion

Limitations

- **Data availability** Although the authors provide the data used in this study, a significant part of it was collected manually. The issue is that exact plane and ship routes on the local scale are usually missing from worldwide datasets.
- **Simplifications made** In this research we do not consider environment-based effects that take place in such region as an Arctic (such as snow storms), that affect the transport routes. These effects are considered as part of temperature restrictions for different transport modes mentioned in the paper.
- **Temperature thresholds** To show rapid changes in transport mode availability due to temperature changes the sigmoid function was used. The threshold value was set to the mean of the sigmoid function to show the transition between seasons. The threshold and decay function could be adjusted for other cases.

Conclusion

As climate change and globalization amplify systemic risks, these frameworks will be indispensable for designing failsafe infrastructure and sustainable ecosystems.

Summary of key findings

The EFN model reveals that Arctic transport networks undergo profound seasonal transformations with immediate implications for infrastructure planning and climate adaptation. Revealed uncertain temperature-dependent catchment areas lead to reconfiguration of flows in service-provider relations in the network. Multilayer interactions underline key settlements in the network across several service dimensions.

Contributions to the field

In real-world applications, nonlinear dynamics in service demand can be observed due to seasonal changes in transport connectivity. As shown in this paper, this effect is most noticeable in a highly climate-dependent region such as the Arctic.

This paper contributes to the field in the following ways:

- The representation of seasonal patterns in the problem of demand satisfaction. Specifically, with the focus on services at the regional (agglomeration) scale.
- Comparative analysis of network dynamics. Seasonal changes in the state of service networks across regions that differ by main types of transport connections, network size and distribution of services (supply) and population (demand).

Policy and planning implications

The data and results obtained allow us to talk about the necessary development of settlements and the definition of their functions. Based on this data, it is possible to draw

conclusions about maintaining and strengthening transport links, as well as developing infrastructure services in settlements. First and foremost, it is proposed that the results be integrated into the process of developing spatial planning programs for Arctic regions. The method is recommended for use as one of the tools for analyzing transport connectivity and the level of infrastructure provision in order to determine strategic development goals and compile a list of necessary measures to maintain the system under study:

- Identification of priority investment projects, decision-making on the allocation of budget funds, and modernization of infrastructure facilities. This will help focus resources where they will be of greatest benefit to the population.
- Development of support programs for the indigenous peoples of the north. Identifying the most vulnerable settlements that need support to preserve their traditional way of life and cultural heritage will ensure their sustainable development.
- Evaluating the effectiveness of existing programs.
- Optimizing logistics. Improving the logistics of delivering goods and materials to remote areas will reduce costs and increase delivery efficiency.
- Developing tourism in remote areas of the north.

Future research directions

Seasonal transitions between network states in the task of meeting urban service needs form the basis for further research into operations dependent on the external climate environment. In particular, the proposed model is the basis for the use of fundamental models involving (influenced by) the external environment. An example of the use of the presented model could be the optimal placement of services, taking into account changes in demand of a known dependence during a given transition period.

Further work on the EFN involves refining the tools and models used. In particular, demand distribution can be specified by more complex behavioral models, including agent-oriented approaches. In addition, the integration of EFN with modern spatio-temporal methods of transport flow analysis (Rahman et al. 2025; Li et al. 2024) and climate-oriented risk assessment and infrastructure adaptation approaches (Rahman et al. 2025; Touloumidis et al. 2025; Burbidge et al. 2024) will allow for a more complete description of the response of regional service networks to changing climatic conditions.

Appendix A: Manual data collection and data manipulations

A.1 Data collection protocols

Data was collected and compiled from various sources to build a multimodal transport network and then analyze the effects of temperature. The process of collecting and calculating time metrics for each type of transport was organized as follows: Motor transport. The initial road network was obtained from OpenStreetMap and via OSMnx Python library (Boeing 2025). A weighted graph was built on the basis of this network. The time cost was calculated using shortest path between 2 nodes on a graph, taking into account the actual length of the segment and the assigned speed, differentiated by seasons: 60 km/h for the summer period and 40 km/h for the winter period. Settlement polygons were represented as their corresponding centroids; then their centroids were mapped to the closest road graph nodes, to make the calculations possible.

Aviation transport. Data on scheduled flights and flight times were collected from official sources: local airline schedules, airport websites, and local government regulations (Aviotryad 2025; Arktika 2025; Yamal 2025; TSAero 2025; Chukotavia 2025; Taimyr 2025). For connected settlements, between which there is no regular communication, or data about it is not given in the schedule, the flight time was estimated based on the geodesic distance, taking into account the typical cruising speed of the most common aircraft, the Mi-8 helicopter.

Water transport. Information about current river and sea routes, as well as travel time, was extracted from the websites of water transport operators (Severnaya Transportnaya Kompaniya 2025). The travel time for water transport was calculated based on the length of the fairway and the accepted average operating speed of 20 km/h. This value is justified by the analysis of the fleet of vessels involved on local routes (boats of the “Yaroslavets” and “KS” types), for which the typical cruising speed in conditions of winding rivers with depth restrictions and the need for frequent stops is 20–25 km/h (Korinevsky and Loginov 2006). This approach provides a conservative and realistic estimate of time costs.

A.2 Validation and evaluation of data quality

The initial cartographic data from OpenStreetMap, although valuable, may be incomplete (Deri et al. 2025) or inaccurate for remote and infrastructurally underdeveloped regions, such as the Arctic (Kwakye et al. 2024). To ensure the completeness and reliability of the transport network model, a step-by-step process of data validation and addition was carried out.: Cross-validation and addition of the route network. The OSM data was compared with alternative cartographic sources (Google 2025). The identified gaps (winter roads, actual roads, and potential ice crossings, etc.) were supplemented based on an analysis of regulations - seasonal resolutions of local authorities on the opening/closing of winter roads and crossings (Republic 2024; Russian 2025; Community 2025), as well as studying current messages in regional social networks reflecting real practices of population displacement.

Connections were evaluated for each type of transport, for which it was possible to obtain the actual time according to the schedule and compare it with the calculated values. This made it possible to characterize the level of detail of the model and outline directions for its future refinement.

Expert evaluation and logical verification: The calculated time matrices for all modes of transport have been logically verified for consistency (for example, flight time cannot exceed the time of a similar car trip over significant distances). Where possible, the results were selectively compared with data from navigation services and expert assessments. The experts in this case were local residents and their relevant experience of real trips.

Appendix B: Comparison logic

B.1 Regional comparison

Selected territories represent a wide range of settlement conditions, infrastructure facilities and transport models. A comparative analysis of key parameters makes it possible to identify systemic differences and common features characteristic of the Arctic as a whole. In total, 9 out of 74 municipalities of the Arctic zone of the Russian Federation were stud-

ied, which displays 1/4 of the territory of the Russian Arctic (Dankin et al. 2022; Pilyasov and Zamyatina 2016; Council 2023).

The comparison reveals a fundamental diversity of spatial and transport models within the considered region: from the relatively connected and infrastructurally developed western periphery (NAO and Mezen Municipal District) through polycentric industrial areas with a river framework (Yamalo-Nenets Autonomous District/Taimyr) to dispersed territories of the east, extremely dependent on seasonal winter routes (Yakutia/Chukotka). Despite these differences, all territories show common systemic features: The critical role of seasonal and informal transport (winter roads, snowmobile trails) complementing the official network. The formation of transport links across administrative borders, dictated by the logic of the natural environment and economic activity. There is a sharp contrast in accessibility between a few hubs (administrative or industrial centers) and a vast periphery. The dependence of the transport framework on a limited number of natural objects (large rivers, coast) and modes of transport (air, winter). Successful adaptation and application of a unified methodology for territories with such different settlement densities, network topology and socio-economic structure will show its universality.

B.2 Service catchments stability thresholds

The identification of unstable settlements and the determination of thresholds after which a settlement is considered “disappeared” in a dynamic model is a projection of the vulnerability of the socio-spatial structure of the region. In the context of the Arctic, the disappearance of the settlement signals a critical degradation of its functions in the transport and social system. Settlement stability thresholds correspond to the following real processes:

1. Loss of critical transport connectivity. A settlement (or group of settlements) ceases to be a stable integrated element of a multimodal network during the season. This leads to a regime of point isolation of the settlement or to the destruction of local socio-economic contours and undermines the basis for the use of limited infrastructure.
2. Cascading effect on neighboring and related settlements. The disappearance of one settlement leads to an increase in the burden on the remaining stable centers, disruption of supply chains and entails long-term socio-demographic reactions (outflow of the most mobile population - youth).
3. A systemic shift in the settlement model. The disappearance of a settlement or an entire cluster of settlements reflects the transition of the system to a qualitatively new state. For example, the transition from a polycentric model (several local support villages) to a monocentric one (absolute dependence on one remote regional center). This dramatically increases the vulnerability of the system as a whole — the failure of one main center paralyzes the entire periphery. Because in a situation where one locality becomes isolated from the entire system, its services cease to be available to the population of other settlements, or services outside its borders become inaccessible to its population.

Thus, the stability threshold we have set is an indicator of the point at which the degradation of transport accessibility ceases to be an operational problem and becomes a threat to the sustainability of the settlement itself as a social unit. This approach makes it possible to use the network model not only to describe the current state, but also to proactively identify “points of failure” in the Arctic settlement system.

Appendix C: Urban policy applications

C.1 Extended discussion

The probabilistic temperature-dependent multilayer network model (EFN) developed in the study allowed us to move from qualitative descriptions to a quantitative analysis of the seasonal dynamics of transport accessibility in the Arctic. The results obtained reveal the systemic properties of Arctic infrastructure networks and suggest new optics for their planning.

Interpretation of key results. Our results show that the external instability hides highly structured patterns of seasonal variability in the system configuration.

1. **Non-linearity and threshold effect.** The model confirmed that the availability of services does not change smoothly, but rather abruptly during periods of seasonal transitions (melting/ice formation) (Ahmed 2005).

The threshold character set by the sigmoid function adequately reflects the reality when a slight warming above the critical point leads to the collapse of the entire network of winter roads, rather than their gradual deterioration.

2. **Vulnerability of polycentricity:** An analysis of the evolution of settlements revealed that the stability of the system depends not on individual roads, but on the stability of local centers of attraction (settlement nodes). The disappearance of such a settlement in the model (the “death” event) means not just the isolation of several settlements, but the disintegration of the local cluster structure, which forces the remaining settlements to reorient to more remote and congested centers. This explains why up to 40% of localities are forced to change service providers: the system is constantly being rebuilt, trying to find a new balance after the loss of local node and new service providers.

3. **Geographical universality of mechanisms with local diversity:** Despite the radical differences between the regions (linear structure of the Northern Administrative District vs. polycentric Yamalo-Nenets Autonomous District vs. dispersed Yakutia/Chukotka), the basic mechanisms of degradation and restoration of connectivity turned out to be common. In all cases, seasonal modes of transport (winter roads, river routes) play a key role, and shifts in accessibility are synchronized with temperature cycles. This proves that the proposed EFN model is precisely the generalizing framework applicable to different configurations of Arctic settlement systems. **Methodology and limits of generalization.** The EFN model contributes by allowing for what has traditionally been left outside the scope of analysis. From static to dynamic: Unlike static infrastructure maps, EFN simulates network performance over time, which is critical for regions where the road exists only 4 months a year.

Integration of the human factor and informal practices: Manual collection of data on unofficial winter roads and local routes made it possible to include real, not only official, travel practices in the model. This increases the adequacy of the model, but at the same time points to its main limitation — dependence on time-consuming collection of expert data. The development of automated methods (track analysis, social media) is a key area for scaling.

Consideration of extreme weather events (blizzards, storms) as part of the generalized temperature threshold, this is a necessary simplification at the current stage. Tactical planning will require embedding probabilistic models of specific hazards that disrupt specific relationships.

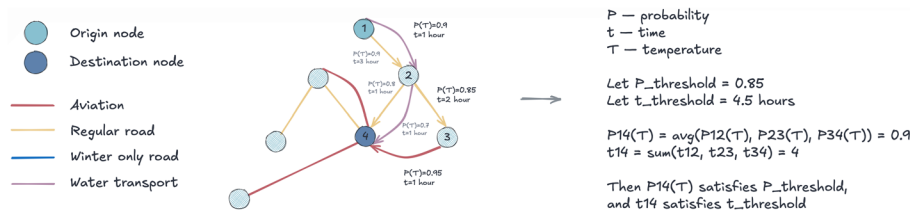


Fig. 13 Network scheme

Accounting for extreme weather events through an aggregated temperature threshold is a strategic simplification adopted in this paper to achieve the main goal of modeling long—term seasonal dynamics. This simplification is based on two key assumptions:

Correlation with seasonality: Peaks in the activity of the main disrupting factors (snow-storms leading to the closure of winter shelters; storms stopping river and sea navigation) are statistically timed to the periods of seasonal transitions, which in the model are determined by temperature thresholds.

Integral effect: On a monthly or seasonal scale, the cumulative effect of multiple short-term extreme events manifests itself as an extension of the period of “unavailability” of a particular type of transport, which is recorded by the model through a shift in the threshold of its operation.

C.2 Formation of a new spatial planning logic

The results obtained are changing the approach to infrastructure management in the Arctic.

1. **Prioritization of investments:** The model allows us to shift the focus from repairing specific road sections to strategically strengthening the backbone links that keep entire communities from disintegrating. Investments in the capitalization of a key winter mine leading to a vulnerable cluster will have a greater systemic effect than repairing several disparate sites.
2. **Adaptive spatial planning:** Identified “accessibility windows” and seasonal patterns of flow redistribution should form the basis for adaptive work schedules for social services, educational and medical teams. Planning of the northern delivery should also take into account not just the navigation period, but the dynamically changing map of accessibility of settlements.
3. **Proactive social risk management:** The fact of seasonal isolation of 15-20% of settlements is not only a transport problem, but also an acute social problem. The EFN model makes it possible to predict these periods and develop proactive measures: the creation of food and medicine supplies, the organization of temporary medical care centers, distance education and consultation schemes (Fig. 13).

Appendix D: Remarks on model formulation

D.1 Failure mode transitions

- Temperature T influences seasonal state S , which is used only to display comparable network states.

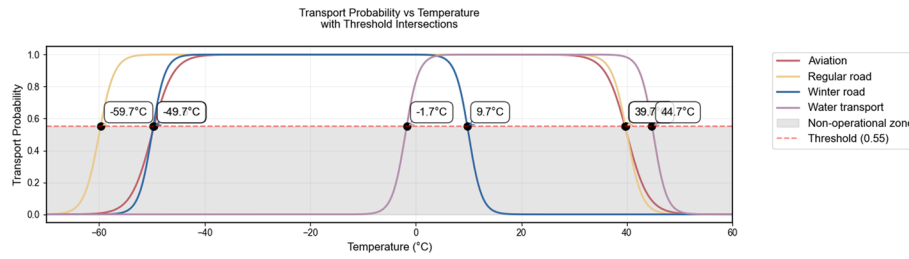


Fig. 14 Temperature-dependent probabilities for different transport modes

Example:

- $P(\text{Water transport}) \rightarrow 0$ when T exceeds operational thresholds.
- $P(\text{Winter road}) \uparrow$ compensates partially if road infrastructure is available.

D.2 Temperature model

Let $T(x, t)$ denote the temperature field at geographic location x and month t . For each settlement i with coordinates x_i , temperature is sampled as

$$T_i(t) := T(x_i, t).$$

Temperature values are used solely to evaluate operational feasibility constraints of transport modes and do not represent disruption, degradation, or any internal state of the settlement nodes.

Because the model operates on an origin–destination abstraction rather than explicit route geometries, we define an effective temperature proxy for each OD pair (i, j) as

$$\tilde{T}_{ij}(t) = g(T_i(t), T_j(t)),$$

where $g(\cdot)$ aggregates endpoint temperatures. In the baseline specification, g is defined as the arithmetic mean of endpoint temperatures; alternative conservative aggregations can be used to reflect stricter feasibility assumptions.

Seasonal effects are not introduced as an explicit categorical variable. Instead, seasonality emerges implicitly through the temporal evolution of $T_i(t)$ and its interaction with mode-specific feasibility thresholds, enabling continuous transitions between accessibility regimes.

For interpretability, we refer to temperature ranges separated by feasibility thresholds as seasonal phases. Transitions between these phases are triggered by crossings of mode-specific temperature thresholds rather than by predefined calendar seasons.

Based on the obtained data, the feasibility of transport mode existence is modeled as a sigmoid function of temperature, reflecting threshold-like operational limits under seasonal climate conditions (Fig. 14).

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Author contributions

G.K. and A.G. wrote the main manuscript; V.T. prepared a part of code implementation and curated the data; S.M. supervised the work.

Data availability

Data and codes for this article are available at the provided github repository: https://github.com/GeorgeKontsevik/seas_on_framed_multilayer_arctic_network_model

Code availability

The source code and data for this article are available in GitHub repository: https://github.com/GeorgeKontsevik/season_framed_multilayer_arctic_network_model

Declarations

Conflict of interest

The authors declare no Conflict of interest.

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References

- Aeroport Yamal (Akyamal): Raspisanie dvizheniya passazhiram: Passenger Flight Schedule. Yamal airport flight schedules for passenger transport networks. (2025). <https://akyamal.ru/passazhiram/raspisanie-dvizheniya/> Accessed 2025–12-08
- Ahmed F (2025) Ice roads and income in remote indigenous communities of Canada. *World Dev Perspect* 37:100666
- Arctic Council of Russia: Arctic Regions of Russia (2023). <https://as.arcticrussia.ru/useful> Accessed 2023–11-20
- Arktika Avialinii (ZAO AO): Services and Operations: Arctic Airlines. Passenger flights, cargo, charters, rescue operations, helicopter maintenance, aviation training complex. Key Arctic carrier. (2025). <https://www.2aoao.ru> Accessed 2025–12-08
- Barrette, P.D., Hori, Y., Kim, A.M.: The canadian winter road infrastructure in a warming climate: Toward resiliency assessment and resource prioritization. *Sustainable and Resilient Infrastructure* (2022) 10.1080/23789689.2022.2094124
- Berthelot G, Tupikina L, Kang M-Y, Sapoval B, Grebenkov DS (2020) Pseudo-darwinian evolution of physical flows in complex networks. *Sci Rep* 10(1):15477
- Boeing G (2025) OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks. GitHub. Software version used: v1.9+ (2025). Python package for downloading and modeling street networks from OpenStreetMap. Essential for urban network analysis.. <https://doi.org/10.5195/geo.2020.76>. <https://geoffboeing.com/publications/osmnx-paper/>
- Burbidge R, Paling C, Dunk RM (2024) A systematic review of adaption to climate change impacts in the aviation sector. *Transp Rev* 44(1):8–33
- Chukotavia Airlines: Article 494: Flight Information. Chukotka regional aviation operations and schedules. (2025). <https://chukotavia.ru/ru/article/view/id/494/> Accessed 2025–12-08
- Coates KS, Broderstad EG (2019) Indigenous peoples of the arctic: Re-taking control of the far north. *The Palgrave handbook of Arctic policy and politics* 9–25
- Dankin MA, Zamyatina NY, Zaitsev AA, Nikitin BV, Poturaeva AV, Ivlieva OD (2022) Reference Settlements of the Russian Arctic. Russia, Scientific Center for Arctic Research (SCRA), Saint Petersburg
- Dankin, M.A., Zamyatina, N.Y., Zaitsev, A.A., Nikitin, B.V., Poturaeva, A.V., Ivlieva, O.D.: Reference Settlements of the Russian Arctic, p. 246. ANO Information and Analytical Center of the State Commission for the Development of the Arctic, Moscow (2022)
- Department of Architecture and Urban Planning under the Head of the Republic of Sakha (Yakutia): Resolution No. 26 On approval of regional standards for urban planning design of the Republic of Sakha (Yakutia). Electronic resource (2022). <https://docs.cntd.ru/document/406031104> Accessed 2024–11-30
- Department of Construction, Housing and Communal Services, Energy and Transport of Nenets Autonomous Okrug: Order No. 4 On approval of regional standards for urban planning design of Nenets Autonomous Okrug. Electronic resource (2024). <https://base.garant.ru/408841477/> Accessed 2024–11-30
- Deri, F., Mara F, Anselmi C (2025) Crowdsourced data for urban planning: A critical evaluation of openstreetmap accuracy and completeness. In: *International Conference on Computational Science and Its Applications*, pp. 403–420. Springer
- Dong Y, Xiao P, Zhang X, Ge D, Yu J, Chen Y, Wu Q, Ma Y, Liu R, Luan W et al (2025) Increased vulnerability of arctic potential ice roads under climate change. *Commun Earth Environ* 6(1):37
- Federal Agency for Technical Regulation and Metrology (Rosstandart): Gost 34066–2017: Snowmobiles - technical requirements and test methods. Technical report (2017)
- Finnish Transport Infrastructure Agency: Winter road maintenance guidelines. Technical report, Finnish Transport Infrastructure Agency (2019). Accessed 14 Dec 2024. <https://vayla.fi/en/maintenance/road-network/winter-maintenance>
- Food and Agriculture Organization (2010) *Climate Variability and Change*. FAO, Rome
- Gangwal U, Siders A, Horney J, Michael HA, Dong S (2023) Critical facility accessibility and road criticality assessment considering flood-induced partial failure. *Sustain Resilient Infrastruct* 8(sup1):337–355
- Google Maps: Google Maps: [Description of map view or search, e.g., Arctic winter roads Yakutia Chukotka] (2025). <https://www.google.com/maps> Accessed 2025–12-08
- Government of the Russian Federation, Ministry of Transport of the Russian Federation: Decree no. 1487 (2020) on the rules of navigation in the northern sea route waters; rules of the russian river register; code of inland waterway transport of the russian federation (federal law no. 24-fz). Technical report (2020). Collection of Russian regulations for inland waterways and Arctic navigation
- Government of Chukotka Autonomous Okrug: Resolution No. 569 On approval of Regional standards for urban planning design of Chukotka Autonomous Okrug. Electronic resource (2011). <https://docs.cntd.ru/document/469504095> Accessed 2024–11-30
- Government of Yamalo-Nenets Autonomous Okrug: Resolution No. 69-P On approval of regional standards for urban planning design of Yamalo-Nenets Autonomous Okrug. Electronic resource (2018). <https://docs.cntd.ru/document/543542347> Accessed 2024–11-30
- Government of Krasnoyarsk Krai: Resolution No. 631-P On approval of regional standards for urban planning design of Krasnoyarsk Krai. Electronic resource (2014). <https://docs.cntd.ru/document/430572436> Accessed 2024–11-30

- Government of Arkhangelsk Oblast: Resolution No. 123-pp On approval of regional standards for urban planning design of Arkhangelsk Oblast. Electronic resource (2016). <https://docs.cntd.ru/document/462627276> Accessed 2024–11–30
- Government of the Republic of Sakha (Yakutia): Information on Active Winter Roads as of January 16, 2024 (2024). <https://xn-80afcrbamb4aave1a.xn-p1ai/press-center/informatsiya-o-dejstvuyushchikh-avtozimmnikakh-po-sostoyaniyu-na-16-01-2024-g> Accessed 2025–12–08
- Gu S, Miao A, Yan D, Hu B, Cheng G, Xu Z (2025) Resilience assessment of urban conventional bus networks in cold regions under snowfall weather. *Reliability Engineering & System Safety* 111824
- Guo X, Du Q, Li Y, Zong X, Bai L (2025) Cascading failure and recovery propagation of metro-bus double-layer network under time-varying passengers. *Transp Res Part D: Transp Environ* 139:104571
- Guo Z, Li Z, Lu C, She J, Zhou Y (2024) Spatio-temporal evolution of resilience: The case of the chengdu-chongqing urban agglomeration in china. *Cities* 153:105226
- Haklay M, Weber P (2008) Openstreetmap: user-generated street maps. *IEEE Pervasive Comput* 7(4):12–18
- Huang L, Wang J, Cheng H (2022) Spatiotemporal changes in ecological network resilience in the shandong peninsula urban agglomeration. *J Clean Prod* 339:130681
- International Maritime Organization: International Code for Ships Operating in Polar Waters (Polar Code). IMO Resolution MSC.385(94). Formal legal code adopted under SOLAS and MARPOL conventions; entered into force 1 Jan 2017 (2014)
- International Maritime Organization: Polar Code: Guidance for Ships Operating in Polar Waters. Operational guidelines and practical implementation of the Polar Code requirements (2017). <https://www.cdn.imo.org/localresources/en/OurWork/Safety/Documents/Polar%20Code%20Guidance.pdf>
- Ji J, Fang L, Chen J, Ding T (2024) A novel framework for urban flood resilience assessment at the urban agglomeration scale. *Int J Disaster Risk Reduct* 108:104519
- Jie R, Wentao Y, Yuting H, Zihao L (2024) Defining urban network resilience: a review. *Front Urban Rural Plan* 2(1):14
- Kiparisov PO, Lagutov VV (2024) Integrated gis-and network-based framework for assessing urban critical infrastructure accessibility and resilience: the case of hurricane michael. [arXiv:2412.13728](https://arxiv.org/abs/2412.13728)
- Kong, X., Doré, G.: Case study of the thermal regime of permafrost underneath the airstrips near hudson bay coast, canada. In: *Cold Regions Engineering 2024: Sustainable and Resilient Engineering Solutions for Changing Cold Regions*, pp. 251–261 (2024)
- Korinevsky PL, Loginov AA (2006) A Brief Description of the Boats of Project 376, 376u. N/A, Moscow, Veliky Novgorod. Technical description of project 376 and 376u boats
- Kwakye D, Marx S, Herfort B, Langer M, Lautenbach S (2024) Data quality of openstreetmap for industrial sites in the arctic. *AGILE GIScience Series* 5:34
- Li J, Li Z, Qi X (2025) A review of multilayer networks-based interregional transportation networks analysis. *Chaos Solitons & Fractals* 192:115993
- Li, Z., Xia, L., Shi, L., Xu, Y., Yin, D., Huang, C.: Opencity: Open spatio-temporal foundation models for traffic prediction. [arXiv:2408.10269](https://arxiv.org/abs/2408.10269) (2024)
- Lowe ME, Sharp S (2021) Gendering human capital development in western alaska. *Econ Anthropol* 8(1):46–60
- Lu H, Zhang C, Jiao L, Wei Y, Zhang Y (2022) Analysis on the spatial-temporal evolution of urban agglomeration resilience: A case study in chengdu-chongqing urban agglomeration, china. *Int J Disaster Risk Reduct* 79:103167
- Mahmoud MR, Roushdi M, Aboelkhear M (2024) Potential benefits of climate change on navigation in the northern sea route by 2050. *Sci Rep* 14(1):2771
- Mei X, Ye W, Li W, Chen C, Li A, Wu J, Du H (2025) Robustness evaluation and optimization of china's multilayer coupled integrated transportation system from a complex network perspective. *Sustainability* 17(16):7398
- Ministry of Regional Development of the Russian Federation, Federal Agency for Technical Regulation and Metrology (Rosstandart), Ministry of Emergency Situations of the Russian Federation: Sp 24.13330.2012 highways; gost r 50992–2019 climatic safety of road vehicles; order no. 731 (2020) on approval of the rules for the use of crossings and pontoon bridges in the russian federation. Technical report (2012). Collection of Russian federal regulations related to highway engineering and operational safety in severe climates
- Ministry of Transport of the Russian Federation: Order No. 128 (31 Jul 2009): Federal Aviation Rules for Preparation and Execution of Flights in Civil Aviation of the Russian Federation. Russian Federal Aviation Regulatory Document (2009)
- Mishina M, Mityagin S, Belyi A, Khruklov A, Sobolevsky S (2024) Towards urban accessibility: Modeling trip distribution to assess the provision of social facilities. *Smart Cities* 7(5):2741–2762
- Mu X, Fang C, Yang Z (2022) Spatio-temporal evolution and dynamic simulation of the urban resilience of beijing-tianjin-hebei urban agglomeration. *J Geog Sci* 32(9):1766–1790
- NASA POWER Project: NASA Prediction Of Worldwide Energy Resources (POWER) Data. NASA Langley Research Center. 2 m air temperature (T2M), spatial resolution 0.5° × 0.625°, accessed via API (2024). <https://power.larc.nasa.gov/>
- Naryan-Marsky Aviotryad (avianao.ru): Raspisanie i rezhim raboty kassy MVL: Local Flight Schedule and Cashier Hours (2025). <https://avianao.ru> Accessed 2025–12–08
- Nordic Road Association (NRF): Nordic arctic highway engineering standards 2020–2024. Technical report, Nordic Road Association (2020). <https://www.nvfnorden.org/>
- Pilyasov AN, Zamyatina NY (2016) Arctic entrepreneurship: Conditions and development opportunities. *Arctic: Ecology and Economics* 4, 4–15
- Povoroznyuk O, Vincent WF, Schweitzer P, Laptander R, Bennett M, Calmels F, Sergeev D, Arp C, Forbes BC, Roy-Léveillé P et al (2022) Arctic roads and railways: social and environmental consequences of transport infrastructure in the circumpolar north. *Arctic Science* 9(2):297–330
- Povoroznyuk O, Vincent WF, Schweitzer P, Laptander R, Bennett M, Calmels F, Sergeev D, Arp C, Forbes BC, Roy-Léveillé P, Walker DA (2023) Arctic roads and railways: Social and environmental consequences of transport infrastructure in the circumpolar north. *Arctic Science* 9(2):297–330. <https://doi.org/10.1139/as-2021-0033>
- Rahman MA, Islam MI, Tabassum M, Bristy IJ (2025) Climate-aware decision intelligence: Integrating environmental risk into infrastructure and supply chain planning. *Saudi J Eng Technol (SJEAT)* 10(9):431–439
- Rouhana F, Jawad D (2025) A spatial-network approach to assessing transportation resilience in disaster-prone urban areas. *ISPRS Int J Geo Inf* 14(7):261

- Russian Federation Ministry of Agency: Regulatory Document No. 469501126 (2025). <https://docs.cntd.ru/document/469501126> Accessed 2025–12-08
- Serova NA, Serova VA (2019) Critical tendencies of transport infrastructure development in the Russian arctic. *Arctic and North* 36:35–46
- Severnaya Transportnaya Kompaniya (STK): Letnyaya navigatsiya: Summer Navigation Schedule, Naryan-Mar Routes (2025). <http://stknao.ru/raspisanie/letnyaya-navigacziya.html> Accessed 2025–12-08
- Stephenson SR, Smith LC, Agnew JA (2011) Divergent long-term trajectories of human access to the arctic. *Nat Clim Chang* 1(3):156–160
- Sun J, Zhai N, Mu H, Miao J, Li W, Li M (2024) Assessment of urban resilience and subsystem coupling coordination in the beijing-tianjin-hebei urban agglomeration. *Sustain Cities Soc* 100:105058
- TSAero: Flight Operations and Schedules. Regional aviation services. (2025). <http://www.tsaero.ru> Accessed 2025–12-08
- Taimyr24.ru: Transport Section: Aviation and Regional Connectivity. Taimyr Peninsula transport news including air connectivity. (2025). https://taimyr24.ru/left_menu/transport/ Accessed 2025–12-08
- Touloumidis D, Madas M, Zeimpekis V, Ayfantopoulou G (2025) Weather-related disruptions in transportation and logistics: A systematic literature review and a policy implementation roadmap. *Logistics* 9(1):32
- Transport Canada: Canadian Aviation Regulations (CAR) Section 602.11: Aircraft Operation in Icing Conditions. Accessed 14 Dec 2024 (2025). <https://laws-lois.justice.gc.ca/eng/regulations/SOR-96-433/>
- Various Aircraft Manufacturers: Operating Instructions for Specific Aircraft Models. (2009). Standard aircraft flight operation manuals for Arctic aviation
- VK Community -67555736 (Yakutia Roads Group): Post on Winter Roads or Arctic Infrastructure Update (2025). https://vk.com/wall-67555736_85497 Accessed 2025–12-08
- Waite T, Evans M, Kholod N, Blahut N, Rowland J (2023) Review of quantitative methods to assess impacts of changing climate and socioeconomic conditions on arctic transportation systems. *Ambio* 52(7):1155–1169
- Wang Y, Cai Y, Xie Y, Chen L, Zhang P (2023) An integrated approach for evaluating dynamics of urban eco-resilience in urban agglomerations of China. *Ecol Ind* 146:109859
- Wang T, Qu Z, Yang Z, Nichol T, Clarke G, Ge Y-E (2020) Climate change research on transportation systems: Climate risks, adaptation and planning. *Transp Res Part D: Transp Environ* 88:102553
- Xinran Wang DW, Shan X (2023) Analysis of regional resilience network from the perspective of relational and dynamic equilibrium. *J Clean Prod*
- Ye Y, Wen H, Zhang L, Zheng Y, Li Y (2025) A multi-factor integrated vulnerability assessment framework for multi-layer intercity road networks considering cross-layer path selection. *Physica A* 130807
- Yin K, Wu J, Wang W, Lee D-H, Wei Y (2023) An integrated resilience assessment model of urban transportation network: A case study of 40 cities in china. *Transp Res Part A Policy Pract* 173:103687
- Yin K, Mostafavi A (2023) Deep learning-driven community resilience rating based on intertwined socio-technical systems features. [arXiv:2311.01661](https://arxiv.org/abs/2311.01661)

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